

SAVE THE DATE · 20 OCTOBER 2026

5th Biosciences Symposium

TU Darmstadt

Connecting ideas. Inspiring discovery. Building collaborations.

Book of Abstracts

Biology Building B2|03, Room 109 · Botanical Garden Campus, TU Darmstadt

Welcome

We are delighted to welcome you to the 5th Biosciences Symposium at TU Darmstadt. This booklet collects the abstracts of every scientific talk, flash talk and poster being presented today, organised by session in the order they appear in the programme.

This symposium is an annual, student-organised event bringing together PhD students, postdocs and principal investigators from across TU Darmstadt's biosciences departments, together with guests from partner institutions. We hope it sparks new conversations, new collaborations and a great day of science.

Programme at a Glance

10:30	**Registration & Opening** Coffee, tea & pastries
11:00	**Welcome Address** Dean, Department of Biology
11:15	**Session I** 4 talks + 5 flash talks + Innovationscouts pitch
12:30	**Snack & Poster Session**
13:30	**Session II** 4 talks
14:30	**Coffee Break & Poster Session**
15:00	**Session III** 3 talks
15:45	**Democratic Voting** Audience votes for Best Talk & Best Poster
15:50	**Group Photo**
16:00	**Awards Ceremony & Closing** Best Talk, Best Poster

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Yunwei Hu — Using deep learning for segmentation and tracking of DSB foci in live cell microscopy

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Arooj Sajjad — Classifying Plant Hinges: Kinematic Engineering Analogies, Terminology, and Diversity
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Lina Dammel — Backcrossing a Resistant and Susceptible C. pomonella Strain to Introgress Resistance
Celine Pana — Differential Impact of Endocrine Therapy on Lysosomal Biogenesis Determines the Response to CDK4/6 Inhibitors in Breast and Prostate Cancer

Poster Session (13 posters)

Ishita Jain — Effect of UV on the mutagenesis and the transcription profile of the Dermal stem cells
Vjosa Berisha — Synechocystis CurT: Functional analysis of post-translational phosphorylated CurT isoforms
Anne-Christin Pohland — Sustainable Carbon Capture through Biomineralisation: Engineering Cyanobacteria for Bicarbonate Accumulation via Extracellular Carbonic Anhydrase

Till Pfeifer — Split-Turbo ID, a sensitive and versatile technique to reveal centromeric protein-protein interactions
Ahmed Bulldan — Cell Type–Selective Targeting by Heterobifunctional Protein Binders via In-cell Enrichment
MohammadSadegh Zaman — Linking Microvascular Architecture to Endothelial Nuclear Organization in a Tumor-Mimicking Model
Arooj Sajjad — Classifying Plant Hinges: Kinematic Engineering Analogies, Terminology, and Diversity
Olga Becker — Synthetic small molecule-binding aptamers as versatile regulatory elements
Alexandra Neradt — Vibrio natriegens as new host for plasmid production
Iurii Sheludko — Nicotiana benthamiana as a competitive host for large-scale oxidative biotransformations and an efficient chassis for metabolic engineering of CYP-mediated biosynthetic pathways
Caroline Erler — Targeting CDK2 in Advanced Prostate Cancer
Daniel Osmar Velazquez — Impact of Cyclin E Overexpression on Prostate Cancer Progression
Jeremie Marlhens — From Genelet Bistability to Programmable Attractor Landscapes

Session I

11:15 – 12:30

MeCP2 interaction networks regulate heterochromatin-dependent higher-order genome organization

Hui Zhang, Postdoc — AG Cardoso

Abstract not submitted.

Modulating Heterocyst Frequency For Increased Hydrogen Production In Anabaena sp. PCC 7120

Patrick Reinmöller, PhD Student — AG Dann

Within heterocysts of the cyanobacterium Anabaena sp. PCC 7120, molecular hydrogen is produced as a waste product during nitrogen fixation (Herrero et al., 2001; Kaneko et al., 2001). Subsequently, hydrogen is metabolised by the uptake hydrogenase (HupSL), making the energy stored in hydrogen available to the heterocysts (Tamagnini et al., 2002). Previous studies showed that an Anabaena sp. PCC 7120 knockout strain lacking the large subunit of the uptake hydrogenase releases up to two times more hydrogen gas than the wild-type strain from its heterocysts (Khetkorn et al., 2020). Meanwhile, heterocyst frequency could be markedly increased by knocking out the patS gene which encodes a heterocyst differentiation signalling peptide suppressing excessive heterocyst differentiation in wildtype filaments (H. S. Yoon & Golden, 2001; H.-S. Yoon & Golden, 1998). Combining both hupSL and patS knock-out mutations represents an interesting approach towards biological hydrogen production for utilization as an energy carrier as a higher heterocyst frequency and increased hydrogen release should increase the hydrogen yields of Anabaena sp. PCC 7120. Our results show that an increase in heterocyst frequency and an increase in hydrogen production can be achieved in patS and hupL3' individual knockout mutant strains, respectively. However, the combination of both genetic modifications unexpectedly results in hydrogen release similar to that of the wild-type strain. Supplementation studies with exogenous carbohydrates through the growth media indicate the unexpectedly low hydrogen yield results from patS hupL3' KO mutants being limited by chemical energy, possibly due to reduced linear electron flow in vegetative cells. Such limitation in photosynthate availability may be overcome through an artificial symbiosis with Azolla filiculoides, a plant that naturally harbours Nostoc azollae as a symbiont.

Intracellular Shotgun Perforation: A Specialized Feeding Strategy of Vampyrellid Amoebae on Tough Algae

Justin Teixeira Pereira Bassiaridis, PhD Student — AG Hess

The rigid cell walls of algae and plants form major structural and diffusional barriers that limit access to the nutrient-rich protoplast. While many heterotrophic protists either extract protoplasm through pre-phagocytotic perforations (“protoplast feeding”) or engulf and digest entire prey cells, little is known about intracellular mechanisms that modify prey walls after phagocytosis. Using live-cell imaging, confocal microscopy and electron microscopy, we found that the vampyrellid amoeba Leptophrys vorax (Vampyrellida, Rhizaria) introduces numerous micron-scale perforations into the cellulose-rich walls of algal prey shortly after phagocytosis. These perforations are accompanied by a transient swelling of the primary food vacuole and precede bulk digestion during the cyst stage. The overall architecture of the cell wall remains largely intact except at these localized lesions. Genome and transcriptome analyses revealed high expression of putative cell wall-degrading enzymes, including glycoside hydrolases targeting cellulose and β-1,3-glucans. Phylogenetic analyses indicate multiple evolutionary origins of these enzymes, including lateral gene transfer from fungi. We propose that this novel feeding strategy, termed “intracellular shotgun perforation”, weakens the diffusion barrier imposed by tough algal walls, thereby facilitating subsequent enzymatic access and intracellular digestion. This previously unrecognized feeding mechanism expands our understanding of how free-living amoebae overcome structurally resistant prey, bridging classical phagocytosis and protoplast feeding.

Using deep learning for segmentation and tracking of DSB foci in live cell microscopy

Yunwei Hu, PhD Student — AG Löwer

Abstract not submitted.

Flash Talks (2 minutes each)

Split-Turbo ID, a sensitive and versatile technique to reveal centromeric protein-protein interactions

Till Pfeifer, Master's Student — AG Heun

Abstract — see Poster Session.

Linking Microvascular Architecture to Endothelial Nuclear Organization in a Tumor-Mimicking Model

MohammadSadegh Zaman, PhD Student — AG Mirzapour

Abnormal tumor vasculature is characterized by altered vessel architecture and disrupted endothelial organization, contributing to impaired tissue perfusion and therapeutic delivery. Understanding how vascular structure relates to cellular organization is therefore important for studying vascular dysfunction. Endothelial nuclear shape and orientation offer a complementary perspective because the nucleus is mechanically connected to the cytoskeleton and responds to its surrounding physical environment. However, the relationship between these nuclear features and local vessel geometry requires further investigation in three-dimensional microvascular networks.

This study examines endothelial nuclear morphology and alignment in self-assembled human microvessels formed within a fibrin–collagen matrix on a microfluidic chip. Static and flow-conditioned networks are investigated under low and tumor-mimicking high concentrations of vascular endothelial growth factor. Quantitative image analysis relates nuclear elongation and orientation to vessel width, length-to-width ratio, curvature and position within the network. The analysis distinguishes local relationships within individual networks from differences between experimental conditions.

By integrating vascular and nuclear morphometry, this work aims to clarify how endothelial nuclear organization relates to microvascular architecture and to establish a basis for future investigations connecting structural features with vascular function.

Classifying Plant Hinges: Kinematic Engineering Analogies, Terminology, and Diversity

Arooj Sajjad, PhD Student — AG Poppinga

Abstract not submitted.

Targeting CDK2 in Advanced Prostate Cancer

Caroline Erler, PhD Student — Uniklinikum Frankfurt

Treatment options for castration-resistant prostate cancer (CRPC) are limited, thus novel therapies are urgently needed. PARP inhibitors (PARPi) have recently been approved and are highly effective in BRCA1/2 mutated CRPCs but patients without BRCA mutation (~ 80%) derive no benefit from treatment. CDK2 is an emerging target in CRPC as it drives resistance to androgen signaling inhibition. CDK2 also mediates replication and promotes homologous recombination (HR) raising the possibility that CDK2 inhibition enhances PARPi efficacy in non-BRCA-deficient patients by inducing replication stress and/or an HR-deficient state. To assess the impact of CDK2i on prostate cancer cells we performed TMT-mass spectrometry. We observed downregulation of several proteins involved in DNA-repair, implying high replication stress, impaired fork stabilization and checkpoint failure in these cells – conditions that sensitize for PARPi. Cotreatment of CDK2i and PARPi indeed showed strong growth inhibition in all prostate cancer models tested. We are further investigating the interplay of CDK2 and DNA-repair to determine patients that could benefit from this combination treatment.

From Genelet Bistability to Programmable Attractor Landscapes

Jeremie Marlhens, PhD Student — Self-Organizing Systems (SOS) Lab

Living cells make binary decisions and maintain them after the triggering signal disappears. Such memory often relies on mutually repressing gene networks, but in cells these circuits are coupled to growth, resource competition, and the surrounding genome.

We build a minimal cell-free version using genelets: short DNA templates transcribed by T7 RNA polymerase whose RNA products repress other templates. Two mutually repressing genelets form a bistable switch. Near its decision boundary, nominally identical reactions can reach different states, allowing us to quantify decision reliability and variability.

We show that local feedback shifts this decision boundary and that coupled switches can be programmed to favor agreement or disagreement despite exclusively repressive interactions. With three switches, network topology restricts the accessible state combinations.

This system provides a minimal, measurable, and rewirable platform for studying molecular decision-making and memory.

Turning research into application-oriented innovation

Innovation Scouts, TU Darmstadt

A 2-minute introduction to the Innovation Scouts, who support TU Darmstadt researchers in turning their research into application-oriented innovations.

InnovationScouts aim to identify application-oriented research at an early stage and provide support where opportunities for knowledge transfer or commercialization arise.

We see ourselves as sparring partners for researchers who wish to better understand the application potential of their research and develop it strategically. We encourage researchers to make active use of our existing scouting services. To this end, we have developed a wide range of formats. Please feel free to contact us at any time, even if your idea is still at an early stage.

Our team includes former researchers from both academic and industrial backgrounds. We bring expertise from the natural sciences and engineering, as well as specialist knowledge in the areas of technology commercialization, intellectual property, and start-up creation.

We will have a stand available for one-to-one discussions throughout the snack and coffee breaks.

Session II

13:30 – 14:30

Coordinated Regulation of Vascular Morphogenesis by Matrix Stiffness and VEGF in a Microfluidic System

Teodora Nazare, PhD Student — AG Mirzapour

Vascular morphogenesis is regulated by the interconnected integration of biochemical and mechanical cues within the cellular microenvironment. Vascular endothelial growth factor is a key regulator of endothelial migration, proliferation, and vascular network formation, while extracellular matrix mechanics regulate cell–matrix adhesion, cytoskeletal tension, mechanotransduction, and vascular organization. On-chip studies have demonstrated that identical Vascular endothelial growth factor gradients can produce distinct sprouting responses depending on matrix density or mechanics. On the other hand, studies using distinct temporal VEGF profiles have shown that vascular morphogenesis benefits from high VEGF levels during early vascular formation, followed by decreasing VEGF concentrations during later stages. Although separate on-chip studies have established methods for directed vascular morphogenesis under varying ECM stiffnesses and VEGF, it remains unclear how VEGF concentration and its effects change in combination with mechanical tuning of the ECM across the successive stages of vasculogenesis, angiogenesis and vascular remodeling. In this study, we used a stromal cell–supported, fibrin-based vasculature on-chip model to modulate ECM stiffness and VEGF availability, while quantitatively following vascular network formation over time using morphometric analysis. We demonstrate that cell number and distance (both for endothelial and fibroblast cells) need to be tuned depending on the stiffness in order to form vascular tubes. In our "normal" (50-20-0.5 ng/ml) VEGF regimen, when the matrix becomes too stiff, the ability of cells to physically deform and remodel the 3D matrix becomes limiting, resulting in slowed down vacuole formation and thinner, poorly connected vessels. Furthermore, we show that a medium VEGF concentration of approximately 20 ng/ml is required in the connection development phase for microvessels to properly maintain growth while reducing sprouting within an 8 mg/ml fibrin hydrogel. These findings demonstrate the importance of considering biochemical and mechanical signals as an integrated and dynamically evolving system and provide a quantitative framework for controlling vascular morphogenesis in engineered tissue models.

Synthetic circular RNAs

Vincent Gunawan, PhD Student — AG Süß

Abstract not submitted.

SeqA beyond replication: A molecular hub for cellular regulation in E. coli

Michał Mozolewski, PhD Student — AG Waldminghaus

The SeqA protein is best known as a regulator of DNA replication initiation in Escherichia coli, where it binds hemimethylated GATC sites to prevent premature re-initiation of chromosome replication. However, accumulating evidence indicates that SeqA performs functions extending far beyond origin sequestration. Cells lacking SeqA exhibit extensive chromosome fragmentation, impaired viability, and strong activation of the SOS response, suggesting a central role in maintaining genome integrity. The molecular mechanisms underlying these phenotypes remain largely unknown. We hypothesize that SeqA acts as a multi-layer coordinator linking chromosome replication, DNA repair, chromosome architecture, and transcriptional regulation to preserve genome stability. A major obstacle in investigating these functions has been the inability to separate the direct effects of SeqA loss from secondary consequences of replication over-initiation and altered chromosomal gene dosage. To overcome this limitation, we developed a novel E. coli experimental system in which chromosome replication proceeds independently of SeqA-mediated origin sequestration through replacement of the native replication origin with the plasmid-derived oriF origin. This system preserves normal replication dynamics while uncoupling SeqA from its classical role in replication-initiation control, allowing its additional cellular functions to be investigated without the confounding effects of replication-associated copy-number bias. Preliminary RNA-seq analyses reveal activation of multiple SOS response genes, including recN, symE, uvrA and umuD, despite comparable replication patterns between the analyzed strains. In addition, synthetic promoter assays demonstrate SeqA-dependent repression of GATC-containing promoters, supporting a previously underappreciated role in transcriptional regulation. Together, these findings suggest that SOS induction in SeqA-deficient cells cannot be explained solely by replication defects and raise the possibility that SeqA directly contributes to the regulation of bacterial DNA damage responses and genomic stability. Elucidating these mechanisms will advance our understanding of how bacteria coordinate DNA replication with genome maintenance and stress adaptation. Given the conservation of the Dam–SeqA system among clinically important Enterobacteriaceae and the growing recognition of the SOS response as a driver of antibiotic resistance evolution, these studies may identify novel targets for antimicrobial strategies aimed at limiting bacterial adaptation rather than bacterial growth.

Releasing Protein Cargo from Synthetic Cells by Coupling Cell-free Expression to Osmotic Actuation

Hongfei Liu, PhD Student — Self-Organizing Systems (SOS) Lab

Macromolecular release from synthetic cells often relies on membrane pores or channels. We explored whether internally generated osmotic stress could increase membrane permeability and promote release. Using double-emulsion microfluidics, we encapsulated an Escherichia coli cell-free expression system in Pluronic L121 polymersomes to co-express the fluorescent protein mScarlet3 and the dextranase PsDex1711 in the presence of dextran. PsDex1711-mediated dextran hydrolysis produced an additional increase in bulk osmolality. Tracking hundreds of individual polymersomes showed that PsDex1711-containing populations expanded, whereas control populations contracted. The distribution of normalized fluorescence revealed a subset with lower values of fluorescence than the main population. In selected expanded polymersomes, integrated internal mScarlet3 fluorescence declined to 18-24% of peak levels, without visible membrane disruption, consistent with protein release. These findings support a proof of concept for using internal gene expression to promote macromolecular release through osmotic actuation without introducing dedicated membrane pores.

Session III

15:00 – 15:45

From Deacylase to Prodrug Activator: Repurposing Sirtuin 2 for Targeted Doxorubicin Release

Yannick Burgdorf, PhD Student — Hochschule Darmstadt (h_da)

Conditional prodrug uncaging offers a strategy to minimize the systemic toxicity of cytotoxic drugs like doxorubicin. We repurpose the intrinsic lysine deacylase activity of human Sirtuin 2 (SIRT2) to serve as an exogenous catalytic trigger for drug release. By installing a benzoyl moiety onto the lysine ε-amine of a cathepsin B-cleavable Phe-Lys peptide, the prodrug remains proteolytically inert. Selective debenzoylation by SIRT2 primes the linker for subsequent protease cleavage, driving self-immolative release of active doxorubicin. Cell assays confirm enzyme-dependent cytotoxicity, showing the validity of this dual-enzymatic approach while ongoing active-site engineering aims at expanding the repertoire of usable masks for the lysine's ε-amine.

Backcrossing a Resistant and Susceptible C. pomonella Strain to Introgress Resistance

Lina Dammel, PhD Student — Julius Kühn-Institut (JKI)

Cydia pomonella (codling moth) is one of the most important insect pests in apple and pear production worldwide. It can be biologically controlled using Cydia pomonella granulovirus (CpGV), which has been used commercially since the late 1980s. However, since 2005, CpGV resistance has been reported in European field populations. So far, five different types of CpGV resistance (termed resistance type I to IV) have been identified, but the genetic basis of this resistance is still not fully understood. This study aims to investigate the genetic basis of type II CpGV resistance using crossing experiments and population genetic analyses. The genomes of the resistant codling moth strain CpR5M.21 and the susceptible CpS strain were sequenced and compared to identify potential genetic markers associated with resistance. To further narrow down the genomic regions involved in resistance, resistant CpR5M.21 males were repeatedly backcrossed with susceptible CpS females for six generations. In each generation, the offspring were exposed to CpGV to select resistant individuals for the next cross. This approach gradually increases the CpS genetic background while retaining genomic regions associated with CpGV resistance. Pool-seq data from different backcross generations are used to compare single nucleotide polymorphism and identify genomic regions associated with type II CpGV resistance.

Differential Impact of Endocrine Therapy on Lysosomal Biogenesis Determines the Response to CDK4/6 Inhibitors in Breast and Prostate Cancer

Celine Pana, PhD Student — Uniklinikum Frankfurt — Molecular Precision Medicine

Abstract not submitted.

Poster Session

On display during Snack & Coffee Break · 13 posters

Effect of UV on the mutagenesis and the transcription profile of the Dermal stem cells

Ishita Jain, PhD Student — AG Cardoso

Abstract not submitted.

Synechocystis CurT: Functional analysis of post-translational phosphorylated CurT isoforms

Vjosa Berisha, PhD Student — AG Dann

The Curvature Thylakoid (CurT) protein is a transmembrane protein found in the thylakoid membranes of cyanobacteria and plant chloroplasts. There it shapes thylakoid membrane architecture by inducing membrane curvature (Armbruster et al., 2013; Luque & Ochoa de Alda, 2014). In Synechocystis sp. PCC 6803, depletion of CurT results in altered thylakoid morphology, impaired photosynthetic performance, and the loss of thylakoid convergence zones, which are contact sites between the thylakoid and the cytoplasmic membrane in Synechocystis (Luque & Ochoa de Alda, 2014; Heinz et al., 2016). In addition to its role in thylakoid organization, CurT depletion causes defects in cell division, showing filamentous cell formation in Synechococcus sp. PCC 7942 and asymmetric cell division in both Synechocystis sp. PCC 6803 and Synechococcus sp. PCC 7942 (Dann et al., 2025). These observations suggest that CurT contributes to multiple membrane-associated cellular processes. One curT gene is encoded in the Synechocystis genome, but four CurT isoforms have been detected in vivo, indicating that post-translational modifications may generate functionally distinct proteins (Spät et al., 2015; Heinz et al., 2016). Among these modifications, phosphorylation and propionylation are proposed to regulate CurT conformation by altering its charge (Spät et al., 2015). To investigate the functional relevance of specific phosphorylation sites, phosphorylatable amino acids can be substituted by residues that either prevent or mimic phosphorylation. Characterizing these phospho-mimetic and phospho-deficient CurT variants may show whether distinct post-translationally modified CurT isoforms differentially regulate thylakoid membrane architecture and cell division.

Sustainable Carbon Capture through Biomineralisation: Engineering Cyanobacteria for Bicarbonate Accumulation via Extracellular Carbonic Anhydrase

Anne-Christin Pohland, Postdoc — AG Dann

Abstract not submitted.

Split-Turbo ID, a sensitive and versatile technique to reveal centromeric protein-protein interactions

Till Pfeifer, Master's Student — AG Heun [Flash Talk]

Abstract not submitted.

Cell Type–Selective Targeting by Heterobifunctional Protein Binders via In-cell Enrichment

Ahmed Bulldan, Postdoc — AG Löwer

Abstract not submitted.

Linking Microvascular Architecture to Endothelial Nuclear Organization in a Tumor-Mimicking Model

MohammadSadegh Zaman, PhD Student — AG Mirzapour

Abnormal tumor vasculature is characterized by altered vessel architecture and disrupted endothelial organization, contributing to impaired tissue perfusion and therapeutic delivery. Understanding how vascular structure relates to cellular organization is therefore important for studying vascular dysfunction. Endothelial nuclear shape and orientation offer a complementary perspective because the nucleus is mechanically connected to the cytoskeleton and responds to its surrounding physical environment. However, the relationship between these nuclear features and local vessel geometry requires further investigation in three-dimensional microvascular networks.

This study examines endothelial nuclear morphology and alignment in self-assembled human microvessels formed within a fibrin–collagen matrix on a microfluidic chip. Static and flow-conditioned networks are investigated under low and tumor-mimicking high concentrations of vascular endothelial growth factor. Quantitative image analysis relates nuclear elongation and orientation to vessel width, length-to-width ratio, curvature and position within the network. The analysis distinguishes local relationships within individual networks from differences between experimental conditions.

By integrating vascular and nuclear morphometry, this work aims to clarify how endothelial nuclear organization relates to microvascular architecture and to establish a basis for future investigations connecting structural features with vascular function.

Classifying Plant Hinges: Kinematic Engineering Analogies, Terminology, and Diversity

Arooj Sajjad, PhD Student — AG Poppinga

Abstract not submitted.

Synthetic small molecule-binding aptamers as versatile regulatory elements

Olga Becker, PhD Student — AG Süß

Abstract not submitted.

Vibrio natriegens as new host for plasmid production

Alexandra Neradt, PhD Student — AG Waldminghaus

Abstract not submitted.

Nicotiana benthamiana as a competitive host for large-scale oxidative biotransformations and an efficient chassis for metabolic engineering of CYP-mediated biosynthetic pathways

Iurii Sheludko, Postdoc — AG Warzecha

Because of their remarkable catalytic diversity, human cytochrome P450 enzymes (CYPs) are valuable tools for biotransformation. They enable regio- and stereospecific oxidation of non-activated hydrocarbon bonds, a process often difficult to achieve by conventional chemical methods. Plants serve as versatile host platforms for expressing foreign CYPs, providing access to cofactor flux and a variety of reducing partners and membrane systems suitable for enzyme integration. However, it is essential to adapt heterologous CYPs to the plant metabolic environment to establish this system as a viable option for large-scale biotechnological applications. To achieve this, we tested various approaches aimed at enhancing the functional activity of heterologous CYPs, including co-expression of heterologous redox partners and adapting CYPs for transfer to various subcellular compartments. While compartmentalization did not enhance catalytic function, we found that the transient co-expression of redox partners significantly influenced the performance of human CYPs in a plant production system. While human cytochrome P450 reductase (CPR) typically enhances or maintains enzyme activity, plant-derived CPRs can either enhance or significantly reduce the metabolism of specific drugs. For most CYP/CPR/substrate combinations, we increased product accumulation by 1.8- to 4.3-fold. By swapping the structural domains of human and plant CPR, we created a hybrid CPR variant that demonstrates improved productivity across all tested CYPs. This variant may serve as a potential universal redox partner for human CYPs expressed in plant hosts. Using scaled transient expression of CYPs and the hybrid CPR gene, we performed preparative biotransformation of the pharmaceutically valuable prodrug yohimbine, yielding around 50 mg of 11-hydroxyyohimbine. This represents the highest ever reported CYP-mediated biotransformation of exogenous substrates in a plant system, comparable to that achieved in yeast. To diversify the spectrum of natural products, we integrated human CYP and CPR enzymes into the engineered irregular monoterpene pathways. These secondary metabolites function as pheromones and insecticides, and demonstrate cytotoxic and antibiotic activities. We established a sustainable platform for irregular monoterpene accumulation in Nicotiana benthamiana, achieving the highest level in plant systems. Two CYPs converted branched and cyclopropane monoterpenes into novel hydroxylated derivatives, with human CPR co-expression aiding in product accumulation. Thus, by integrating human CYPs into a network of heterologous genes from different organisms, we initiated new biosynthetic pathways in N. benthamiana and expanded the range of irregular monoterpene structures. Acknowledgements: YVS acknowledges the financial support from the German Research Foundation (DFG), grant number 516587177.

Targeting CDK2 in Advanced Prostate Cancer

Caroline Erler, PhD Student — Uniklinikum Frankfurt

Treatment options for castration-resistant prostate cancer (CRPC) are limited, thus novel therapies are urgently needed. PARP inhibitors (PARPi) have recently been approved and are highly effective in BRCA1/2 mutated CRPCs but patients without BRCA mutation (~ 80%) derive no benefit from treatment. CDK2 is an emerging target in CRPC as it drives resistance to androgen signaling inhibition. CDK2 also mediates replication and promotes homologous recombination (HR) raising the possibility that CDK2 inhibition enhances PARPi efficacy in non-BRCA-deficient patients by inducing replication stress and/or an HR-deficient state. To assess the impact of CDK2i on prostate cancer cells we performed TMT-mass spectrometry. We observed downregulation of several proteins involved in DNA-repair, implying high replication stress, impaired fork stabilization and checkpoint failure in these cells – conditions that sensitize for PARPi. Cotreatment of CDK2i and PARPi indeed showed strong growth inhibition in all prostate cancer models tested. We are further investigating the interplay of CDK2 and DNA-repair to determine patients that could benefit from this combination treatment.

Impact of Cyclin E Overexpression on Prostate Cancer Progression

Daniel Osmar Velazquez, PhD Student — Uniklinikum Frankfurt

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Thank you for joining us.

5th Biosciences Symposium · TU Darmstadt · 20 October 2026

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